

Microstructure and Properties of Aluminum-Containing Refractory High-Entropy Alloys

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A new metallurgical strategy, high-entropy alloying (HEA), was used to explore new composition and phase spaces in the development of new refractory alloys with reduced densities and improved properties. Combining Mo, Ta, and Hf with “low-density” refractory elements (Nb, V, and Zr) and with Ti and Al produced six new refractory HEAs with densities ranging from 6.9 g/cm³ to 9.1 g/cm³. Three alloys have single-phase disordered body-centered cubic (bcc) crystal structures and three other alloys contain two bcc nanophases with very close lattice parameters. The alloys have high hardness, in the range from $H_v = 4.0$ GPa to 5.8 GPa, and compression yield strength, $\sigma_{0.2} = 1280$ MPa to 2035 MPa, depending on the composition. Some of these refractory HEAs show considerably improved high temperature strengths relative to advanced Ni-based superalloys. Compressive ductility of all the alloys is limited at room temperature, but it improves significantly at 800°C and 1000°C.

INTRODUCTION

Recently, several refractory high-entropy alloys (HEAs) were produced and their microstructures and properties were reported.^{1–8} Some of these alloys exhibit high-temperature properties that are competitive with the current Ni-based superalloys. In particular, refractory HEAs have the potential to operate at much higher temperatures; for example, some have considerably higher strengths at temperatures above 800°C than Ni superalloys. Unfortunately, the balance of properties such as their high density, low room-temperature (RT) ductility and insufficient oxidation resistance are issues that still need to be resolved by subsequent metallurgical studies.

Historically, the alloying element Al is associated with several beneficial properties in many high-temperature (HT) alloys. For example, in Ni- and Ni/Fe-based superalloys Al forms coherent γ' (Ni₃Al-based) particles, which improve the strength and creep resistance.⁹ Al also forms a protective oxide scale, which improves oxidation and corrosion resistance. In steels, Al is widely used as a deoxidizer, ferrite stabilizer, and a precipitation-forming element, and it is the most effective alloying element for preventing austenite grain growth.^{10,11}

Finally, in Co-Cr-Fe-Ni-based HEAs, Al stabilizes a body-centered cubic (bcc, ferrite) phase and considerably increases strength.^{12–14} In all these cases, Al addition reduces the alloy density.

In this work, Al was added to several refractory HEAs to reduce alloy density, increase HT strength and specific strength, and improve oxidation resistance. An additional objective of this work was to study whether Al additions stimulate or suppress formation of intermetallic (IM) phases in refractory HEAs. All reported refractory HEAs that do not contain Cr are bcc structures, whereas alloys with Cr contain Laves phases and have a considerably decreased ductility at temperatures below 800°C.^{5,8} The introduction of the Laves phases seems to be associated with the atomic radius of Cr ($r_{Cr} = 128$ pm), which is much smaller than that of other refractory elements (~ 146 pm). Although Al forms a number of IM phases with transition elements in conventional alloys, it favors the formation of disordered bcc phases in Co-Cr-Fe-Ni-based HEAs.^{12,15,16} It is worth noting that the atomic radius of Al ($r_{Al} = 143$ pm) is similar to the atomic radii of refractory elements. One may therefore expect that addition of Al may impede formation of IM phases in HEAs that do not contain Cr and thus may improve ductility.

EXPERIMENTAL PROCEDURES

Six high entropy alloys containing different refractory elements, as well as titanium and aluminum, were prepared by vacuum arc melting nominal mixtures of the corresponding elements. The purities of Al, Hf, Mo, Nb, Ta, Ti, V, and Zr were 99.999%, 99.9%, 99.99%, 99.95%, 99.9%, 99.98%, 99.9%, and 99.95%, respectively. Respective mixtures of the elements were arc melted inside a water-cooled copper cavity. High-purity molten titanium was used as a getter for residual oxygen, nitrogen, and hydrogen. To achieve a homogeneous distribution of elements in the alloys, each alloy was remelted five times, was flipped for each melt, and was in a liquid state for about 5 min during each melting event. The prepared specimens were about 12 mm high, 30 mm wide, and 100 mm long. The actual alloy compositions determined with the use of an electron-probe microanalyzer (EPMA) are given in Table I.

The alloy A1 ($\text{AlMo}_{0.5}\text{NbTa}_{0.5}\text{TiZr}$) and A3 ($\text{Al}_{0.4}\text{Hf}_{0.6}\text{NbTaTiZr}$) are modifications of the earlier reported $\text{CrMo}_{0.5}\text{NbTa}_{0.5}\text{TiZr}$ ⁵ and HfNbTaTiZr ³ alloys, respectively. Alloy A2 ($\text{AlNb}_{1.5}\text{Ta}_{0.5}\text{Ti}_{1.5}\text{Zr}_{0.5}$) is a modification of alloy A1, where Mo is replaced with Nb and half of Zr is replaced with Ti to further reduce the density. Alloy A4 ($\text{Al}_{0.3}\text{NbTaTi}_{1.4}\text{Zr}_{1.3}$) is produced from alloy A3 by replacement of heavy Hf with lighter Ti and Zr. Alloy A5 ($\text{Al}_{0.3}\text{NbTa}_{0.8}\text{Ti}_{1.4}\text{V}_{0.2}\text{Zr}_{1.3}$) is a modified A4 alloy, where some amount of Ta is replaced with lighter V. Finally, partial substitution of Zr with Al (in alloy A5) produced alloy A6 ($\text{Al}_{0.5}\text{NbTa}_{0.8}\text{Ti}_{1.5}\text{V}_{0.2}\text{Zr}$). After solidification, the alloys A1 and A2 were hot isostatically pressed (HIP) at 1400°C and 207 MPa for 2 h, and then annealed at 1400°C for 24 h in continuously flowing high-purity argon. The alloys A3 to A6 were HIP at 1200°C and 207 MPa for 2 h, and then annealed at 1200°C for 24 h in continuously flowing high-purity argon. During HIP and annealing, the samples were covered with Ta foil to minimize oxidation. The cooling rate after annealing was 10°C/min.

The crystal structure was identified with the use of an x-ray diffractometer, Cu K α radiation, and the 2 Θ scattering range of 10° to 140°. The alloy densities were measured with an AccuPyc 1330 V1.03 helium pycnometer (Micromeritics Instrument Corporation, Norcross, GA). Vickers microhardness

was measured on polished cross-section surfaces using a 136° Vickers diamond pyramid under 500-g load applied for 20 s. The microstructure was analyzed with a scanning electron microscope (SEM) Quanta 600F (FEI Company, Hillsboro, OR) equipped with backscatter electron (BSE), energy-dispersive x-ray spectroscopy (EDS), and electron backscatter diffraction (EBSD) detectors.

Compression tests of rectangular specimens with the dimensions of $\sim 4.7 \text{ mm} \times 4.7 \text{ mm} \times 7.7 \text{ mm}$ were conducted using a computer-controlled Instron (Instron Corporation, Norwood, MA) mechanical testing machine outfitted with a Brew vacuum furnace and silicon carbide dies. A constant ramp speed that corresponded to an initial strain rate of 10^{-3} s^{-1} was used. The deformation of all specimens was video recorded and image correlation software Vic-Gauge (Correlated Solutions, Inc., Columbia, SC) was used to register strain versus stress curves.

RESULTS

Density, Microhardness, and Crystal Structure

The densities of the produced refractory alloys are given in Table II. They vary from the minimum value of $\rho = 6.88 \text{ g/cm}^3$ for alloy A2 to the maximum value of $\rho = 9.05 \text{ g/cm}^3$ for alloy A3. Using the densities of pure elements and the alloy compositions (Table I), the densities ρ_{mix} of disordered solid solutions of the same compositions are estimated using the rule of mixtures:

$$\rho_{\text{mix}} = \frac{\sum c_i A_i}{\sum \frac{c_i A_i}{\rho_i}} \quad (1)$$

where c_i , A_i , and ρ_i are the concentration, atomic weight, and density of element i . The calculated ρ_{mix} values are very close to the densities of homogenized alloys (see Table II). Similar observations have been previously reported for other high-entropy alloys.^{1,3,5}

Vickers microhardness values are given in Table II. Alloy A1 has the maximum microhardness of $H_v = 5.8 \text{ GPa}$ and alloy A2 has the minimum $H_v = 4.0 \text{ GPa}$. Four other alloys have H_v of approximately 5.0 GPa. Comparisons of the properties of the Al-containing refractory HEAs with those of

Table I. Chemical compositions (in at.%) of the alloys studied in this work

Alloy	ID	Al	Hf	Mo	Nb	Ta	Ti	V	Zr
$\text{AlMo}_{0.5}\text{NbTa}_{0.5}\text{TiZr}$	A1	20.4	0	10.5	22.4	10.1	17.8	0	18.8
$\text{AlNb}_{1.5}\text{Ta}_{0.5}\text{Ti}_{1.5}\text{Zr}_{0.5}$	A2	21.9	0	0	32	9	26.7	0	10.3
$\text{Al}_{0.4}\text{Hf}_{0.6}\text{NbTaTiZr}$	A3	7.9	12.8	0	23	16.8	18.9	0	20.6
$\text{Al}_{0.3}\text{NbTaTi}_{1.4}\text{Zr}_{1.3}$	A4	5.7	0	0	23.5	17.6	27.2	0	26
$\text{Al}_{0.3}\text{NbTa}_{0.8}\text{Ti}_{1.4}\text{V}_{0.2}\text{Zr}_{1.3}$	A5	5.2	0	0	23.4	13.2	27.7	4.3	26.2
$\text{Al}_{0.5}\text{NbTa}_{0.8}\text{Ti}_{1.5}\text{V}_{0.2}\text{Zr}$	A6	11.3	0	0	22.3	13.1	27.9	4.5	20.9

Table II. Density [experimental (Exp.) and calculated using Eq. 1 (Calc.)] and microhardness of the produced alloys

Alloy	ID	Density (Exp.) (g/cm ³)	Density (Calc.) (g/cm ³)	Microhardness (GPa)
AlMo _{0.5} NbTa _{0.5} TiZr	A1	7.40 ± 0.08	7.3 ± 0.1	5.8 ± 0.1
AlNb _{1.5} Ta _{0.5} Ti _{1.5} Zr _{0.5}	A2	6.88 ± 0.07	6.8 ± 0.1	4.0 ± 0.1
Al _{0.4} Hf _{0.6} NbTaTiZr	A3	9.05 ± 0.05	9.0 ± 0.1	4.9 ± 0.1
Al _{0.3} NbTaTi _{1.4} Zr _{1.3}	A4	8.18 ± 0.05	8.1 ± 0.1	4.8 ± 0.1
Al _{0.3} NbTa _{0.8} Ti _{1.4} V _{0.2} Zr _{1.3}	A5	7.78 ± 0.05	7.6 ± 0.1	4.9 ± 0.1
Al _{0.5} NbTa _{0.8} Ti _{1.5} V _{0.2} Zr	A6	7.42 ± 0.05	7.3 ± 0.1	5.2 ± 0.1

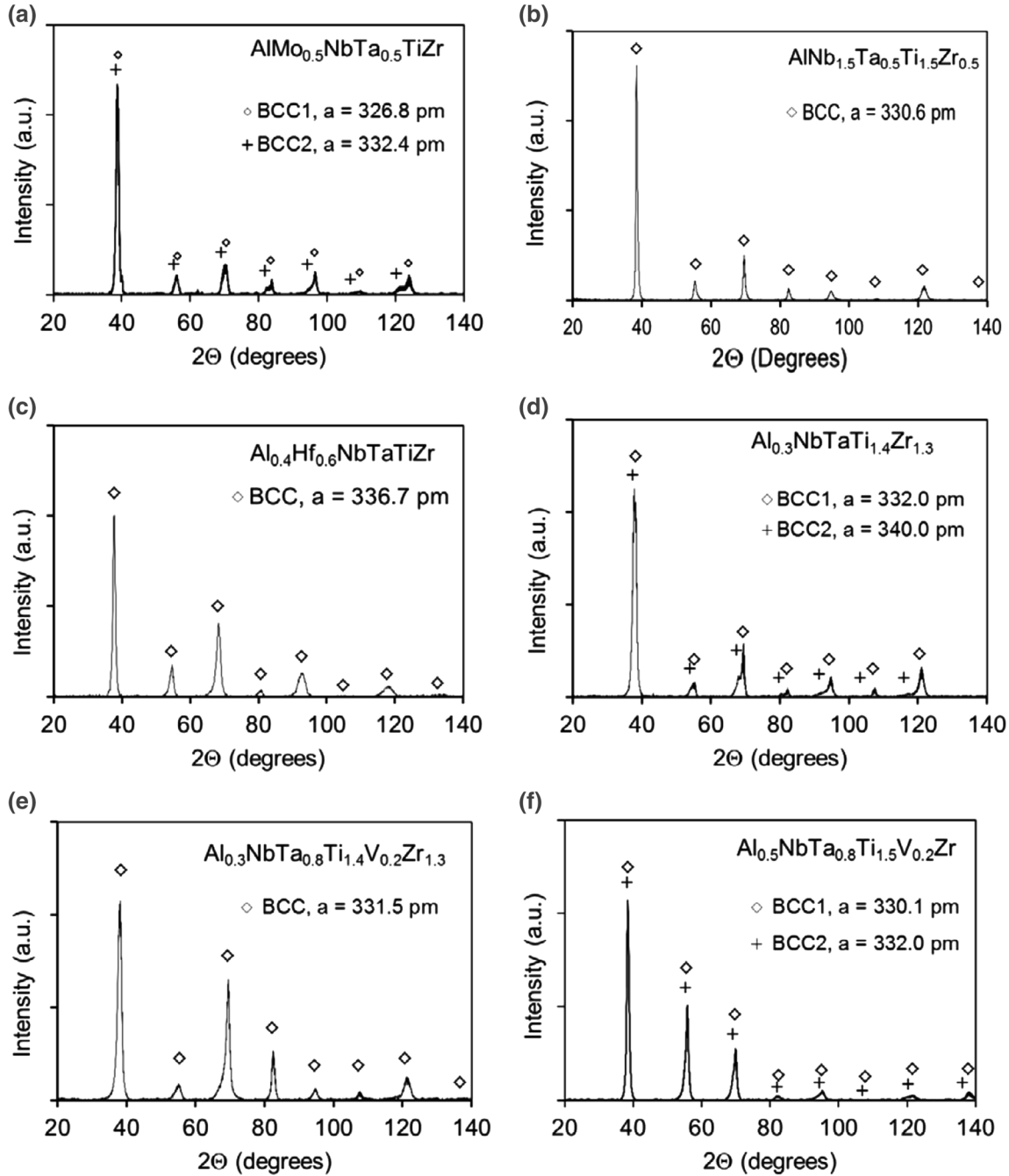
**Fig. 1. X-ray diffraction patterns of the cast and homogenized alloys (a) A1, (b) A2, (c) A3, (d) A4, (e) A5, and (f) A6 (see Table I for the alloy composition).**

Table III. Lattice parameters of BCC phases in cast alloys

Alloy	ID	Phase ID	a (pm) ^a	a (pm) ^b
AlMo _{0.5} NbTa _{0.5} TiZr	A1	BCC1	326.8	325.9
		BCC2	332.4	332.2
AlNb _{1.5} Ta _{0.5} Ti _{1.5} Zr _{0.5}	A2	BCC	330.6	330.3
Al _{0.4} Hf _{0.6} NbTaTiZr	A3	BCC	336.7	337.2
Al _{0.3} NbTaTi _{1.4} Zr _{1.3}	A4	BCC1	332.0	332.0
		BCC2	340.0	340.0
Al _{0.3} NbTa _{0.8} Ti _{1.4} V _{0.2} Zr _{1.3}	A5	BCC	331.5	332.3
Al _{0.5} NbTa _{0.8} Ti _{1.5} V _{0.2} Zr	A6	BCC1	330.1	330.6
		BCC2	332.0	

^aAfter annealing ^bAfter 50% compressive deformation at 1000°C.

previously reported refractory HEAs clearly show the beneficial effects from alloying with Al. In particular, replacement of Cr in CrMo_{0.5}NbTa_{0.5}TiZr⁵ with Al results in a considerable decrease in the alloy density, from 8.23 g/cm³ to 7.40 g/cm³, and a noticeable increase in microhardness from 5.3 GPa to 5.8 GPa. Substitution of 0.4 Hf in HfNbTaTiZr³ with Al decreases the density from 9.94 g/cm³ to 9.05 g/cm³ and increases H_v from 3.8 GPa to 4.9 GPa. The hardness of these alloys is similar to the hardness of high-strength steel.¹⁷

X-ray diffraction patterns of the produced alloys in the annealed condition are shown in Fig. 1. Bragg peaks show the presence of only bcc phases. According to x-ray analysis, A2, A3, and A5 are single phase alloys, and A1, A4, and A6 are the two-phase alloys. No superlattice peaks are observed, which suggest that the phases may be disordered. The lattice parameters of the identified phases are given in Table III.

Microstructure of Heat-Treated Alloys

After HIP and annealing, all the alloys have equiaxed grain structure (Fig. 2). The average grain size is 75 μ m, ~2000 μ m, 140 μ m, 200 μ m, 180 μ m, and 100 μ m in alloys A1, A2, A3, A4, A5, and A6, respectively. Subgrain boundaries with misorientation angles less than 1.5° inside grains and large second-phase particles at grain boundaries are present in alloy A1 (Fig. 2). Higher magnification images reveal the presence of very fine, basket-weave, nanolamellar structures inside the grains of two-phase alloys A1, A4, and A6 (Fig. 3). A closer look reveals discrete cuboids aligned along crystallographic cube directions analogous to a rafted microstructure in Ni-based superalloys.⁹ This basket-weave nanostructure consists of two phases and the lamellar spacing is estimated to be ~70 nm, 40 nm, and less than 15 nm in alloys A1, A4, and A6, respectively. An EBSD analysis confirms that these two phases have bcc crystal structures and the same crystallographic orientations inside every grain. The nanoparticles coarsen at grain and sub-

grain boundaries and reveal two phases with distinct Z contrasts (as in A4, Fig. 3). The darker phase consists mainly of lighter elements and the brighter phase contains larger amounts of heavy elements. An EDS analysis shows that the dark nanoparticles are enriched with Al and Zr, whereas the bright particles are enriched with Nb (Table IV).

Higher magnification images also reveal precipitation of secondary phases at grain boundaries of alloys A2, A3, and A5 (Fig. 3). In A2, the second phase is present as a continuous, ~220-nm-thick layer enriched with Zr and Al (Table IV). In A3 and A5, grain boundaries are decorated by bright and dark nanoparticles. The composition of bright nanoparticles is close to the composition of the matrix, whereas dark nanoparticles are enriched with Zr.

Compression Properties

Figure 4 shows engineering stress versus engineering strain curves for the studied refractory alloys. The compression properties of these alloys, such as yield strength $\sigma_{0.2}$, peak strength σ_p , and fracture strain δ are given in Table V.

The alloys have high $\sigma_{0.2}$ and σ_p at room temperature (RT): $\sigma_{0.2}$ varies from a minimum value of 1280 MPa for A2 to the highest value of 2035 MPa for A6. σ_p is minimum for A2 (1367 MPa) and maximum for A1 (2368 MPa). The RT compression ductility δ is 10% for A1 and A3, but it is only 5.0% or less for other alloys. With an increase in the temperature, $\sigma_{0.2}$ and σ_p decrease, while δ increases (Table V). After heating from RT to 800°C, $\sigma_{0.2}$ decreases the most rapidly, by ~82%, in alloy A4, while alloy A1 is the most heat resistant and loses only 20% of its strength. $\sigma_{0.2}$ of alloys A3, A5, and A6 decreases by ~60% and that of A2 by 43%. In the temperature range from 800°C to 1000°C, A4 is the most heat-resistant alloy, showing ~35% drop in $\sigma_{0.2}$, while alloys A5 and A6 show the highest, 72–76% decrease in $\sigma_{0.2}$. At 1000°C, $\sigma_{0.2}$ varies from 166 MPa for A5 to 745 MPa for A1. The significant

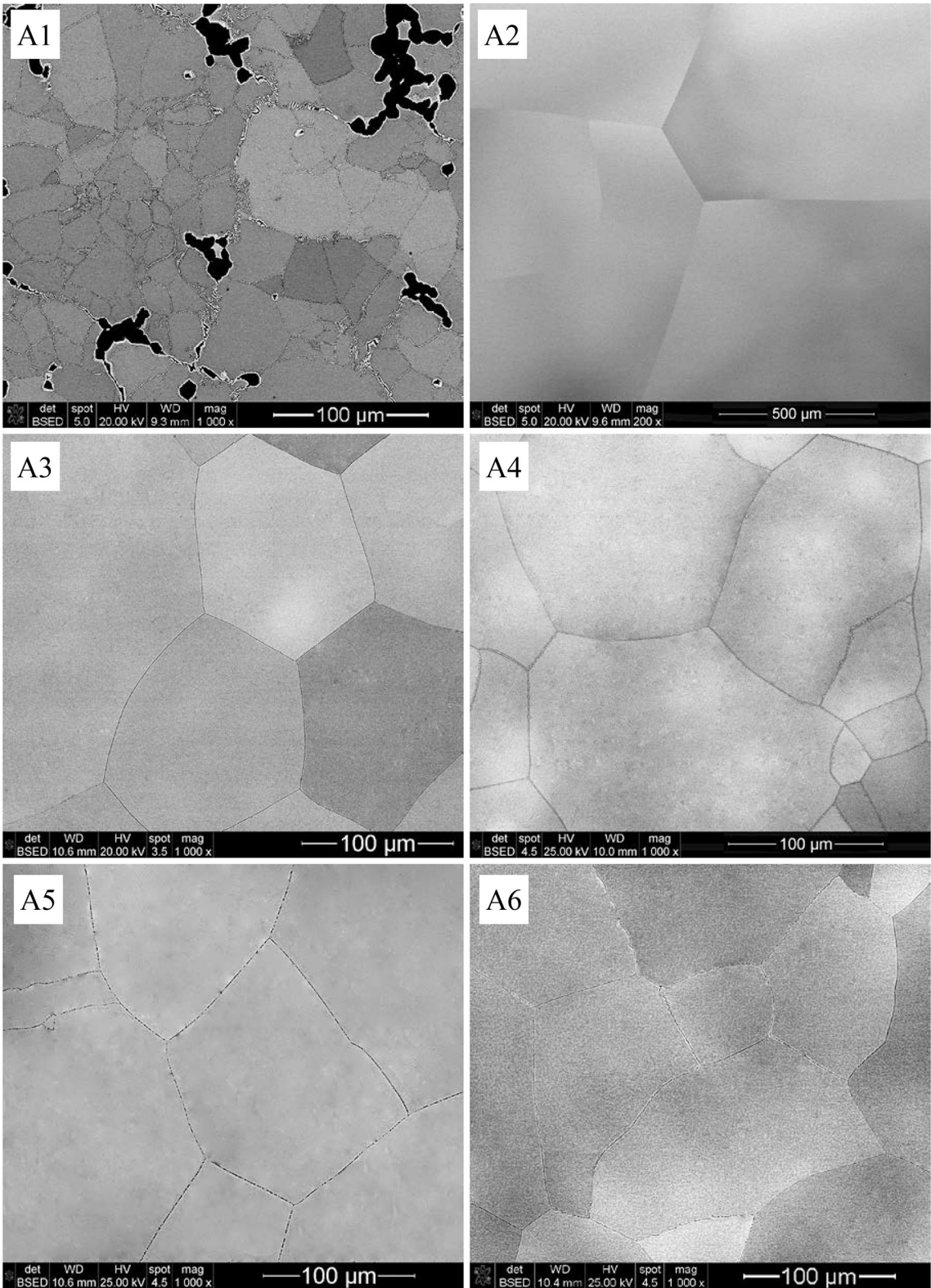


Fig. 2. Backscatter electron images of the microstructure of the refractory HEAs: A1, A2, A3, A4, A5, and A6.

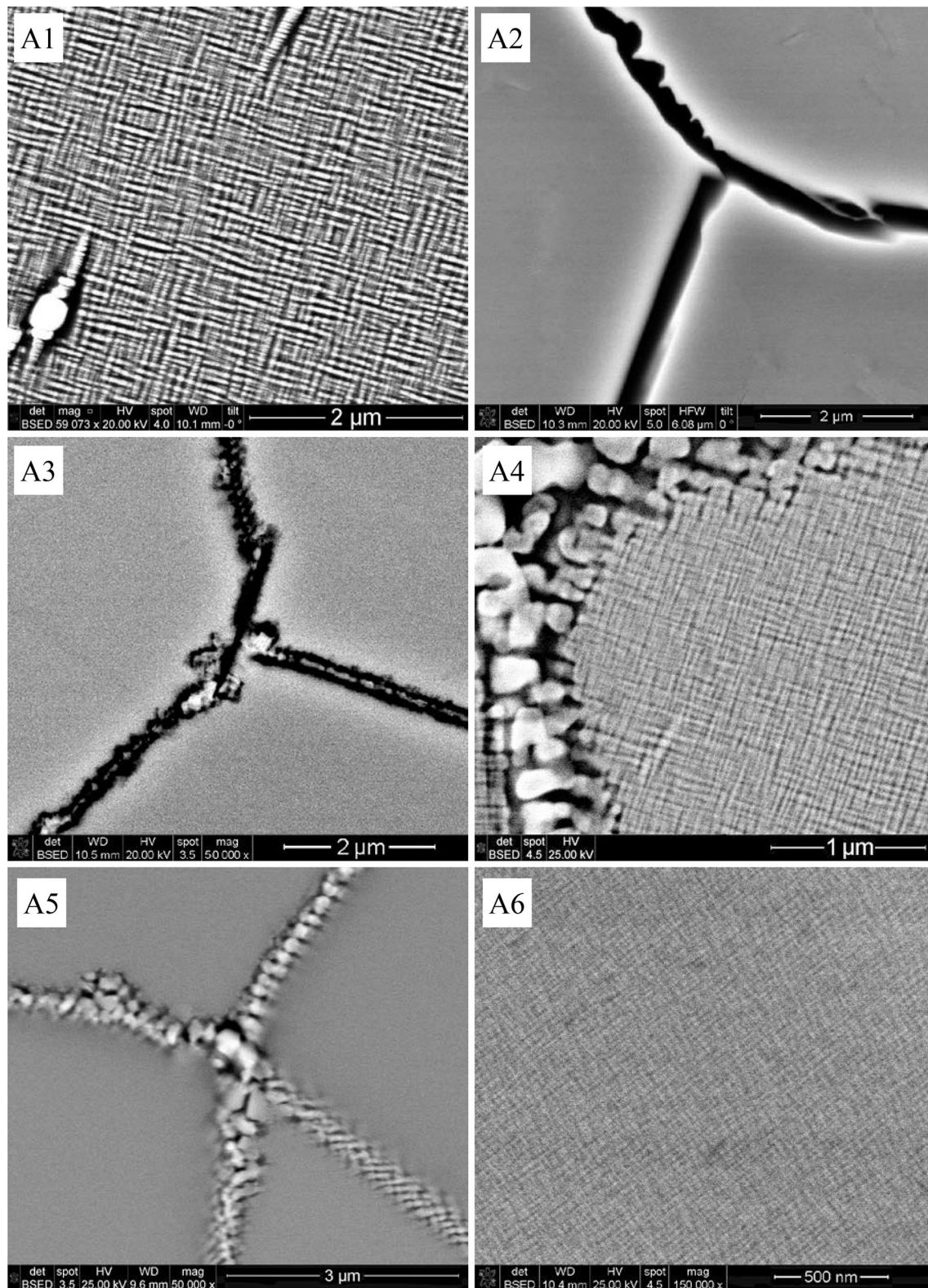


Fig. 3. Fine microstructure of grain interior and grain boundaries of the A1–A6 HEAs.

Table IV. Chemical composition (in at.%) of constituents present in different alloys

At.%	Al	Hf	Mo	Nb	Ta	Ti	V	Zr
<i>Alloy A1 (AlMo_{0.5}NbTa_{0.5}TiZr)</i>								
Matrix grains	19.9		11.3	22.9	10.6	18.5		16.8
Large particles at GBs	39.5		0.0	10.9	2.3	7.2		40.1
Bright nanoparticles	15.4		14.5	25.6	12.8	19.2		12.5
Dark nanoparticles	27.6		6.8	19.1	7.2	15.2		24.1
<i>Alloy A2 (AlNb_{1.5}Ta_{0.5}Ti_{1.5}Zr_{0.5})</i>								
Matrix grains	21.7			32.2	9.1	26.9		10.1
Dark layer at GBs	30.0			23.1	5.9	17.8		23.2
<i>Alloy A3 (Al_{0.4}Hf_{0.6}NbTaTiZr)</i>								
Matrix grains	7.9	12.8		23.0	16.8	18.9		20.6
<i>Alloy A4 (Al_{0.3}NbTaTi_{1.4}Zr_{1.3})</i>								
Matrix grains	5.9			22.9	16.8	27.7		26.7
Bright nanoparticles at GBs	5.2			24.4	18.2	26.3		25.8
Dark nanoparticles at GBs	6.2			19.4	14.7	27.6		31.9
<i>Alloy A5 (Al_{0.3}NbTa_{0.8}Ti_{1.4}V_{0.2}Zr_{1.3})</i>								
Matrix grains	5.1			22.2	12.6	27.7	2.5	27.8
Bright nanoparticles at GBs	5.2			22.5	12.3	26.5	4.6	28.9
Dark nanoparticles at GBs	6.3			18.4	10.7	27.7	4.5	32.4
<i>Alloy A6 (Al_{0.5}NbTa_{0.8}Ti_{1.5}V_{0.2}Zr)</i>								
Matrix grains	11.4			22.0	13.1	28.2	4.7	20.6
Dark nanoparticles at GBs	23.8			15.4	8.4	19.2	3.3	29.9

decreases in strength with increasing temperature may result from the onset of thermally activated flow above room temperature, as it is typical for bcc metals. In addition, alloys A1, A4, and A6 may soften from the dissolution of one of the two bcc phases at elevated temperatures. All the alloys, except A1, have high compression ductility ($\delta > 50\%$) at 800°C and 1000°C. However, alloy A1 shows only 11% ductility at 800°C.

Figure 5 shows a comparison of the specific yield strength ($\text{SYS} = \sigma_{0.2}/\rho$) of the studied HEAs and two representative Ni-based superalloys, wrought Inconel 718¹⁸ and cast Mar-M247.^{19,20} Although the SYS values of the HEAs represent compression properties in the annealed condition, the SYS values of the Ni superalloys reflect tensile properties after typical heat treatment. RT SYS of all six HEAs are considerably higher than SYS of the superalloys. At 800°C, SYS of all the studied HEAs, except A4, is higher than that of IN718, and three HEAs, A1, A2 and A6 are stronger than MAR-M247. At 1000°C, all six HEAs are stronger than IN718, and two alloys A1 and A2 are stronger than Mar-M247. Moreover, the SYS of A1 is more than twice higher than that of Mar-M247 at 1000°C.

Microstructure and Phases After 50% Compression Deformation at 1000°C

X-ray diffraction analysis showed that deformed samples of alloys A1–A5 have the same phases with almost the same lattice parameters, as before deformation (Table III). However, only one bcc

phase was identified by x-ray in alloy A6, which had two bcc phases before deformation.

BSE images of the deformed microstructures are shown in Figs. 6 and 7. The matrix grains of all alloys are elongated in the directions of local plastic flow, which are inclined by $\sim 90^\circ$ to 60° to the compression direction and deformation bands crossing some grain boundaries can be observed (see Figs. 6, A3 and 7, A5). Characteristic bands and spots with different contrasts, similar to those shown in low-magnification images of alloys A2 (Fig. 6) and A6 (Fig. 7), are present inside deformed grains in alloys A2, A3, A4, and A6. These distinctive contrasts are likely caused by different electron channeling conditions and indicate the presence of a subgrain structure and internal stresses, which lead to slight misorientations of regions inside the deformed grains.²¹ The subgrain structure is clearly identified on high-magnification images (see Figs. 6, A2 and 7, A6). Fine recrystallized grains are formed at the grain boundaries of deformed grains in alloys A3, A4, and A5.

Second-phase particles remain present at grain boundaries of alloys A1, A2, and A6. The basket-weave nanophase structure inside the grains of alloys A1 and A4 becomes coarser and a characteristic relief forms inside the grains of these two alloys (see Figs. 6, A1 and 7, A4). This relief is likely a result of the interaction of the local material flow with the interface boundaries between the nanolamellar phases. Fine second-phase particles are present inside deformed grains of alloy A6 (Fig. 7).

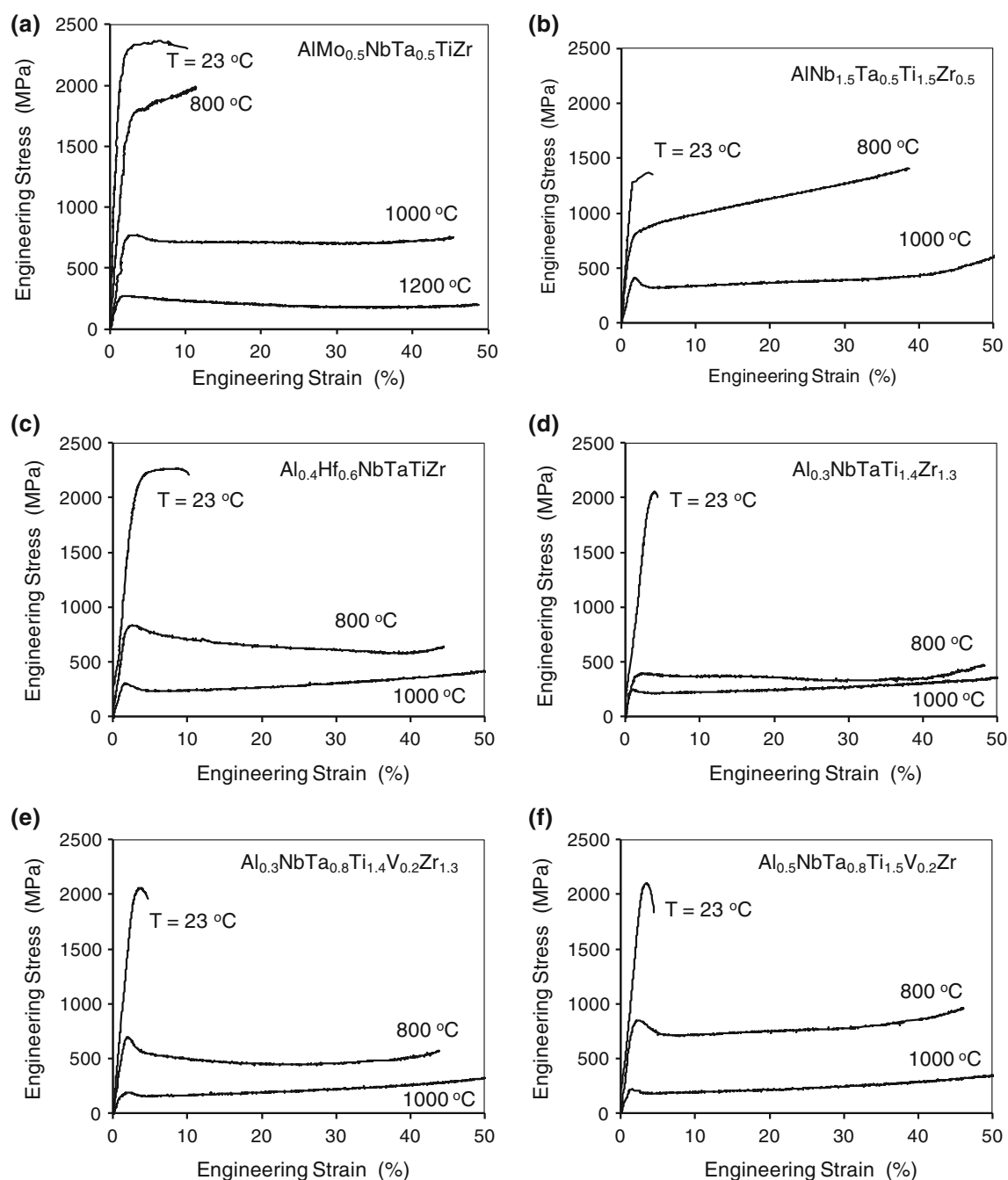


Fig. 4. Engineering stress-strain deformation curves of six high-entropy alloys.

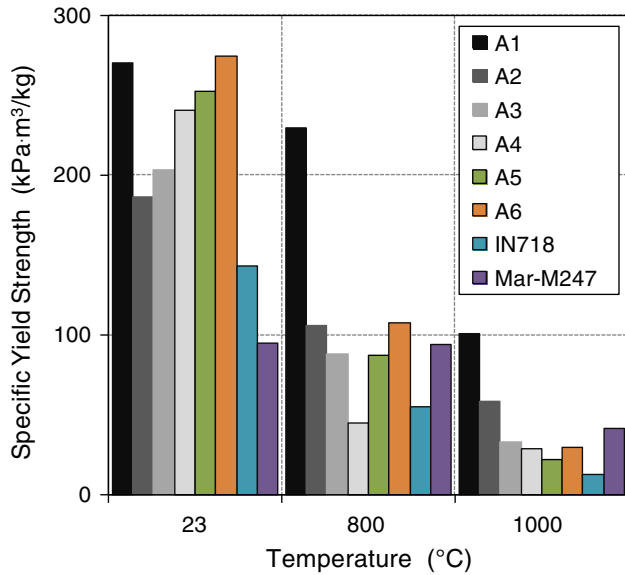
DISCUSSION

The addition of Al to the refractory HEAs considered in this work has produced several beneficial effects over the earlier reported refractory HEAs. First, the Al-containing HEAs have considerably reduced density relative to their parent alloys. For example, partial (40%) substitution of Hf in HfNbTaTiZr ($\rho = 9.94 \text{ g/cm}^3$ ³³) with Al produces a 9% lighter $\text{Al}_{0.4}\text{Hf}_{0.6}\text{NbTaTiZr}$ (A3) alloy, while complete substitution of Cr with Al in $\text{CrMo}_{0.5}\text{NbTa}_{0.5}\text{TiZr}$

($\rho = 8.23 \text{ g/cm}^3$ ³⁵) produces the $\text{AlMo}_{0.5}\text{NbTa}_{0.5}\text{TiZr}$ (A1) alloy with $\rho = 7.40 \text{ g/cm}^3$. Second, the RT hardness and strength of five Al-containing refractory HEAs are noticeably higher than the respective properties of the earlier reported HEAs.^{1,3,5,7} For example, the RT hardness and yield strength of A1 are 9.4% and 12.7% higher than those of $\text{CrMo}_{0.5}\text{NbTa}_{0.5}\text{TiZr}$. The RT hardness and yield strength of A3 are 28.9% and 84.8% higher than the respective properties of HfNbTaTiZr. The exceptionally high strength of A1, A4, and A6 alloys is

Table V. Compression yield strength $\sigma_{0.2}$, maximum strength σ_p , and fracture strain δ of the studied alloys at different temperatures

Alloy ID/properties	A1	A2	A3	A4	A5	A6
$T = 25^\circ\text{C}$						
$\sigma_{0.2}$ (MPa)	2000	1280	1841	1965	1965	2035
σ_p (MPa)	2368	1367	2269	2054	2061	2105
δ (%)	10	3.5	10	5.0	5.0	4.5
$T = 800^\circ\text{C}$						
$\sigma_{0.2}$ (MPa)	1597	728	796	362	678	796
σ_p (MPa)	1810	1000	834	397	693	851
δ (%)	11	> 12	> 50	> 50	> 50	> 50
$T = 1000^\circ\text{C}$						
$\sigma_{0.2}$ (MPa)	745	403	298	236	166	220
σ_p (MPa)	772	415	455	247	189	225
δ (%)	> 50	> 50	> 50	> 50	> 50	> 50

**Fig. 5. Comparison of the specific yield strength of six high-entropy alloys (compression) and two superalloys, IN718¹⁸ and Mar-M247^{19,20} (tension).**

probably caused by a nano-basket-weave structure of two bcc phases. These phases have different compositions but the same crystal structures (bcc) and the same crystallographic orientations inside every grain. Moreover, they have similar lattice parameters. For example, the lattice mismatch is 1.7%, 2.4%, and 0.58% for A1, A4, and A6 alloys, respectively. These observations suggest that the interfaces between these two phases in the nano-basket-weave structure are coherent or semicoherent. Because of the high thermal stability of the two-phase nanostructure, A1 retains its high strength at temperatures up to 1200°C and show compressive strengths that are superior to tensile strengths in some advanced Ni-based superalloys. The three other Al-containing HEAs, A2, A3, and A5, have a single-phase bcc crystal structure and their strengths are

consistent with the refractory bcc metals and a hybrid form of high-concentration solid-solution-strengthening mechanism.^{3,22} Although the alloys show good workability at elevated temperatures, their room-temperature ductility needs to be improved.

Another advantage from the addition of Al and avoiding Cr in refractory HEAs is suppression of the formation of brittle, topologically close-packed Laves phases, which improves RT ductility (e.g., from 5% for CrMo_{0.5}NbTa_{0.5}TiZr to 10% for Al-Mo_{0.5}NbTa_{0.5}TiZr). A Laves phase is generally formed in an alloy containing two types of atoms with the atom size ratio close to $\sqrt{3}/2 \approx 1.225$.²³ Among the refractory elements, Cr has the smallest atomic radius and forms binary Laves phases with Nb, Ta, Ti, and Zr.²⁴ At the same time, the atomic radius of Al is very close to the atomic radii of other elements in refractory HEAs. A small atomic size difference between alloying elements has recently been shown to be a necessary criterion favoring the formation of disordered solid solutions and discouraging formation of intermetallic phases in high-entropy alloys.^{25–28} On the other hand, Al has a different crystal structure and forms a number of binary intermetallic phases with refractory elements.^{22,24} Therefore, it is worth understanding why the refractory high-entropy alloys containing Al do not seem to form intermetallic phases. It would be convenient to reference the multicomponent phase diagrams for these chemistries to make this assessment. However, at this time it is not apparent that enough binary and ternary reference measurements have been documented in the literature and integrated into a CALPHAD scheme to make this a realistic approach. There has been an emerging set of heuristic criteria for suppressing the intermetallics phases in HEAs.

Zhang et al.^{26,29} have recently defined two parameters, δ_r and Ω , to predict the composition range of solid-solution phase formation in HEAs. The atomic size difference parameter δ_r is calculated using the following equation:

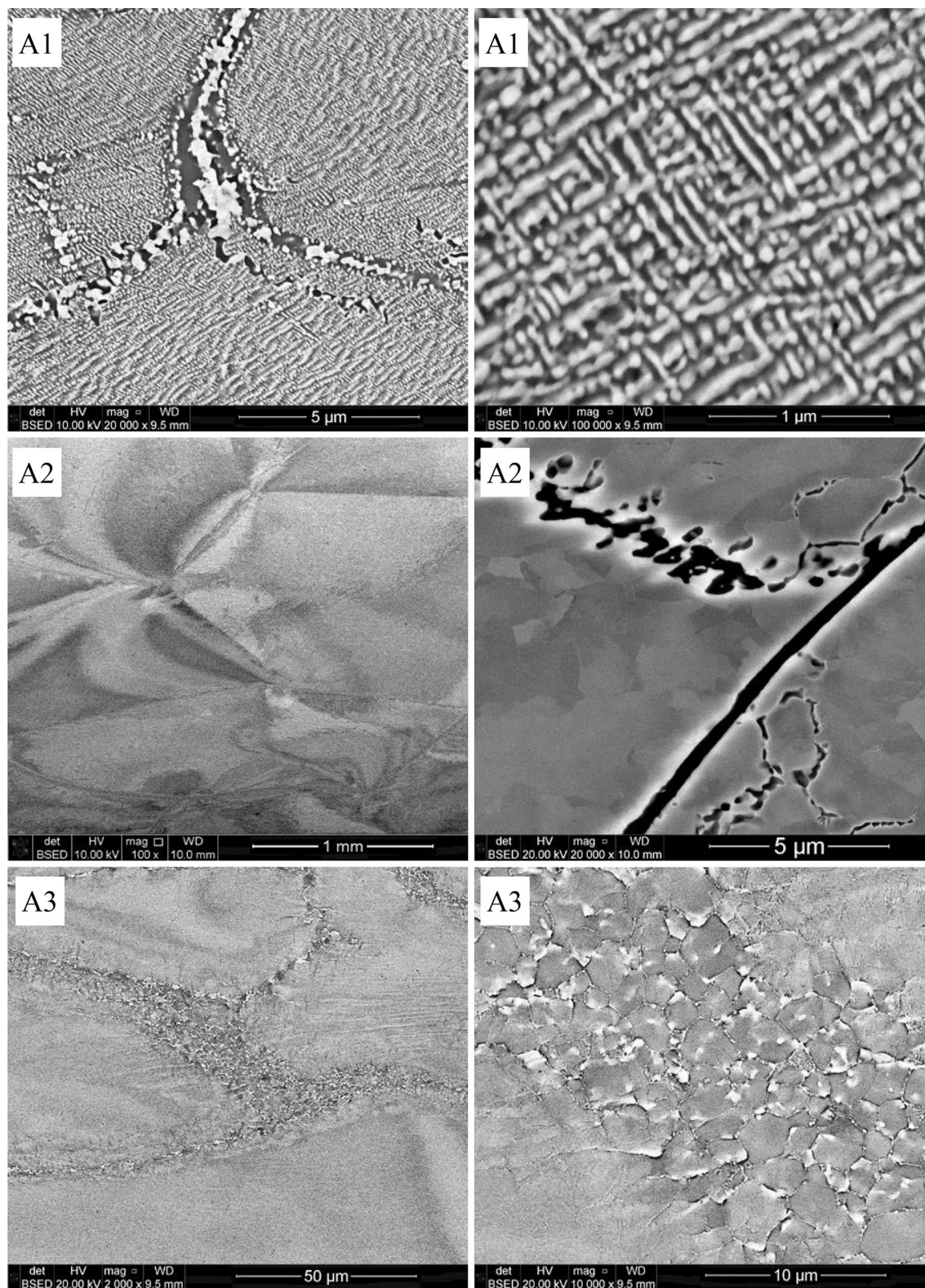


Fig. 6. Microstructure of the refractory HEAs, A1, A2, and A3 after 50% compression deformation at 1000°C.

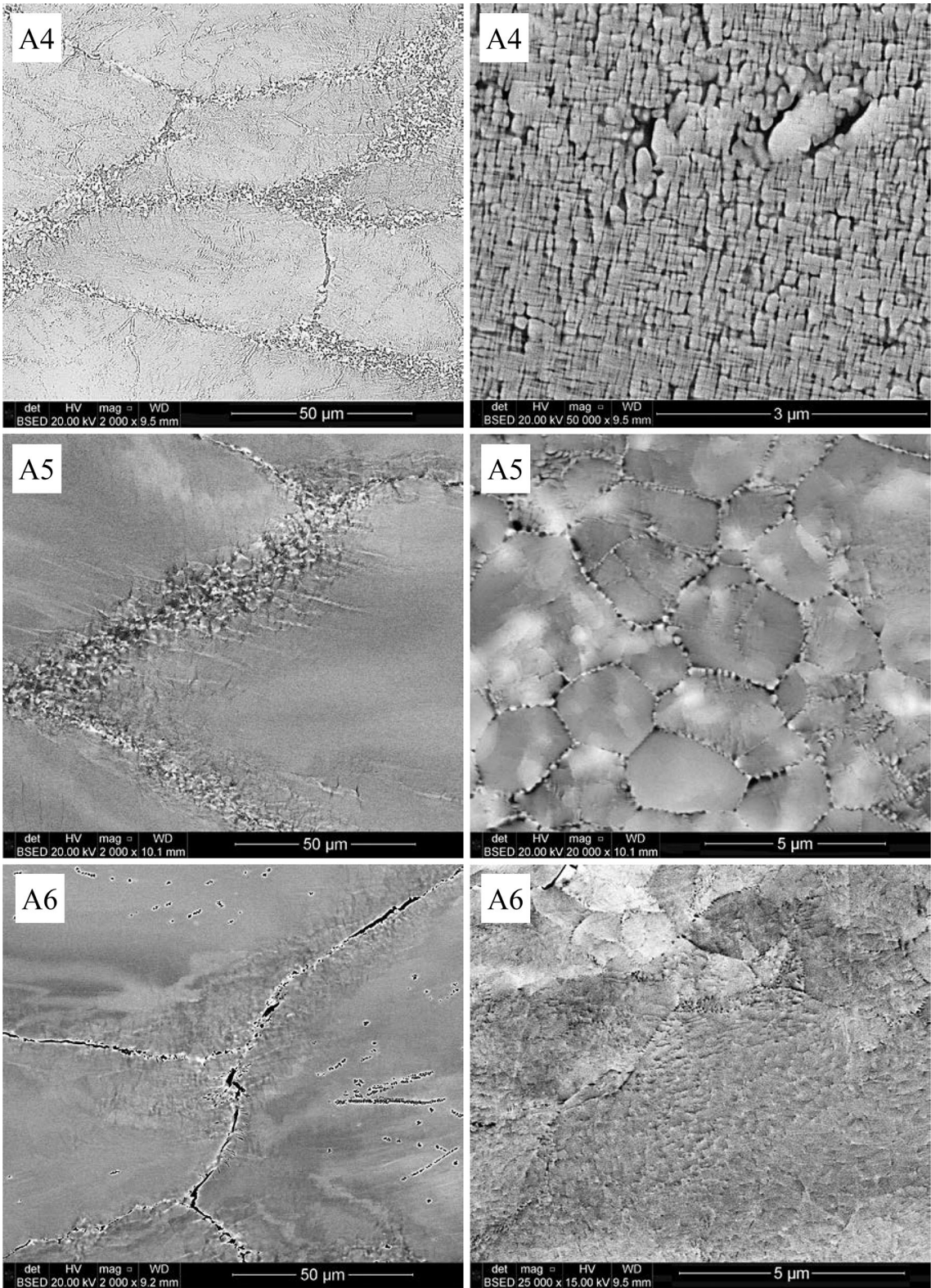


Fig. 7. Microstructure of the refractory HEAs, A4, A5, and A6 after 50% compression deformation at 1000°C.

Table VI. Calculated enthalpy and entropy of mixing, average melting temperature T_m , atomic radius difference δ_r , and parameter Ω for the studied and earlier reported refractory HEAs

ID/Ref.	Composition	Phase content	ΔH_{mix} (kJ/mol)	ΔS_{mix} (J/mol/K)	T_m (K)	δ_r (%)	Ω
A1	AlMo _{0.5} NbTa _{0.5} TiZr	2BCC	−16.6	14.5	2188	4.70	1.92
A2	AlNb _{1.5} Ta _{0.5} Ti _{1.5} Zr _{0.5}	BCC	−16.1	12.5	2121	3.99	1.65
A3	Al _{0.4} Hf _{0.6} NbTaTiZr	BCC	−6.2	14.5	2385	4.25	5.58
A4	Al _{0.3} NbTaTi _{1.4} Zr _{1.3}	2BCC	−4.0	12.6	2360	4.29	7.43
A5	Al _{0.3} NbTa _{0.8} Ti _{1.4} V _{0.2} Zr _{1.3}	BCC	−4.0	13.3	2315	4.80	7.80
A6	Al _{0.5} NbTa _{0.8} Ti _{1.5} V _{0.2} Zr	2BCC	−9.8	13.9	2234	4.81	3.18
1,2	MoNbTaW	BCC	−6.6	11.5	3158	5.37	5.49
1,2	MoNbTaVW	BCC	−4.7	13.4	2963	5.70	8.46
3,4	HfNbTaTiZr	BCC	2.7	13.4	2523	3.93	12.71
5	CrMo _{0.5} NbTa _{0.5} TiZr	2BCC+Laves	−5.0	14.5	2418	6.79	7.07
7,8	NbTiVZr	BCC	−0.2	11.5	2250	6.13	148.21
7,8	CrNbTiZr	BCC+Laves	−5.3	11.5	2250	7.13	4.94
7,8	CrNbTiVZr	BCC+Laves	−4.7	13.4	2236	7.17	6.36
7,8	NbTiV ₂ Zr	3BCC	−1.1	13.4	2237	6.50	26.35

$$\delta_r = 100 \% \sqrt{\sum c_i (1 - r_i/\bar{r})^2} \quad (2)$$

where c_i is the atomic fraction of element i in the alloy, $\bar{r} = \sum c_i r_i$ is the average atomic radius, and r_i is the atomic radius of element i . The parameter Ω takes into account the combined effects of the mixing entropy, $S_{\text{mix}} = -R \sum c_i \ln c_i$, mixing enthalpy, $H_{\text{mix}} = \sum 4\omega_{ij}c_i c_j$, and effective melting temperature $T_m = \sum c_i T_{mi}$ of an HEA and is defined as:

$$\Omega = \frac{T_m \Delta S_{\text{mix}}}{|\Delta H_{\text{mix}}|} \quad (3)$$

where ω_{ij} is a concentration-dependent interaction parameter between elements i and j in a subregular solid-solution model³⁰ and T_{mi} is the melting temperature of element i . By analyzing a number of nonrefractory HEAs, Yang and Zhang²⁹ have found that the formation of intermetallic phases in a HEA in the as-cast state generally does not occur when $\Omega > 1.1$ and $\delta_r < 6.6\%$.

Using the known atomic radii and melting temperatures of the elements, as well as ω_{ij} values from Ref 30, the parameters ΔH_{mix} , ΔS_{mix} , T_m , Ω , and δ_r were calculated for the studied, as well as for earlier reported, refractory HEAs, and the results are given in Table VI. Although ΔS_{mix} of the studied alloys differs not much, from the minimum value of 12.5 J/mol/K for A2 to the maximum one of 14.5 J/mol/K for A1, ΔH_{mix} is very sensitive to alloy composition. ΔH_{mix} has a maximum value of −4.0 kJ/mol for alloys A4 and A5 and a minimum value of −16.6 kJ/mol for A1. The A1, A2, A3, and A6 alloys have considerably higher negative values of ΔH_{mix} than previously reported refractory HEAs, including those containing intermetallic phases (see Table VI). These results show that large negative values of ΔH_{mix} do not trigger the formation of intermetallic phases in these Al-containing refractory alloys.

All the studied alloys have $\Omega > 1.1$ and $\delta_r < 6.6\%$, which favor the formation of solid-solution phases. Previously reported refractory HEAs, even those containing intermetallic phases, also have $\Omega > 1.1$ (see Table VI). However, only solid-solution forming HEAs have $\delta_r < 6.6\%$, while alloys containing intermetallic phases have $\delta_r > 6.6\%$. One can therefore conclude that the Ω – δ_r criterion by Yang and Zhang²⁹ predicts the formation of solid-solution phases in the present refractory high-entropy alloys very well.

CONCLUSION

1. The phase composition, microstructure, and properties of six new refractory high-entropy alloys are reported. In addition to the refractory elements, the alloys also contain Al and Ti, which reduce the alloy density.
2. Three alloys, AlMo_{0.5}NbTa_{0.5}TiZr (A1), Al_{0.3}NbTaTi_{1.4}Zr_{1.3} (A4), and Al_{0.5}NbTa_{0.8}Ti_{1.5}V_{0.2}Zr (A6) contain two bcc phases with very similar lattice parameters. Both phases are present in the form of interpenetrating nanolamellae creating basket-weave nanostructure inside grains. One phase is enriched with Nb and another phase is enriched with Zr. The densities of these alloys are 7.4 g/cm³ (A1), 8.2 g/cm³ (A4), and 7.4 g/cm³ (A6), and their Vickers hardness is 5.8 GPa, 4.8 GPa, and 5.2 GPa, respectively.
3. Three other alloys, AlNb_{1.5}Ta_{0.5}Ti_{1.5}Zr_{0.5} (A2), Al_{0.4}Hf_{0.6}NbTaTiZr (A3), and Al_{0.3}NbTa_{0.8}Ti_{1.4}V_{0.2}Zr_{1.3} (A5) are essentially single-phase bcc alloys. However, nanometer-sized secondary phases are present at grain boundaries. The densities of these alloys are 6.9 g/cm³ (A2), 9.0 g/cm³ (A3), and 7.8 g/cm³ (A5), and their Vickers hardness is 4.0 GPa, 4.9 GPa, and 4.9 GPa, respectively.

4. Compression tests reveal very high strength of some of the studied alloys in the temperature range from 23°C to 1000°C, superior to the tensile strengths Ni-based superalloys, such as Inconel 718 and Mar-M247. Compressive ductility is limited at 23°C, suggesting that little or no tensile ductility may result at this temperature.

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